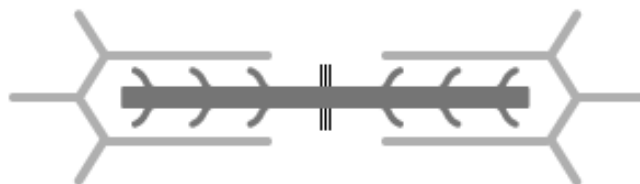


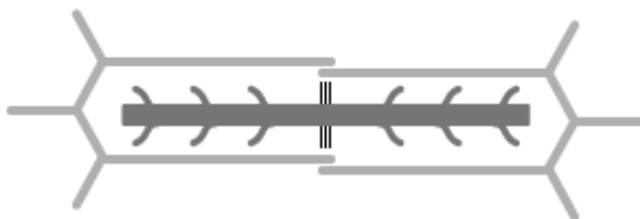
A skeletal muscle sarcomere is shown above. The ability of a resting sarcomere to contract is dependent on the existence of an overlap between actin and myosin filaments and the presence of a gap between the ends of the actin filaments and the M-line. Given this, which of the following diagrams best represents a resting sarcomere with the potential to generate the greatest amount of contractile force?

✓ ☒ A.



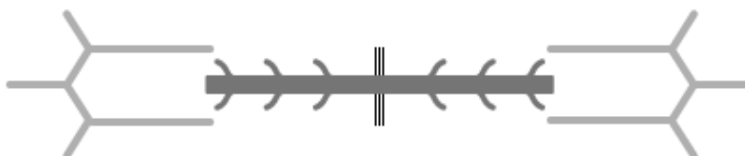
[50%]

☐ B.



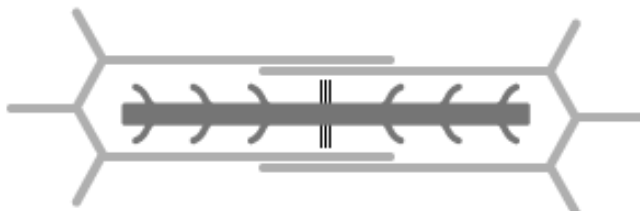
[6%]

☐ C.



[28%]

☐ D.

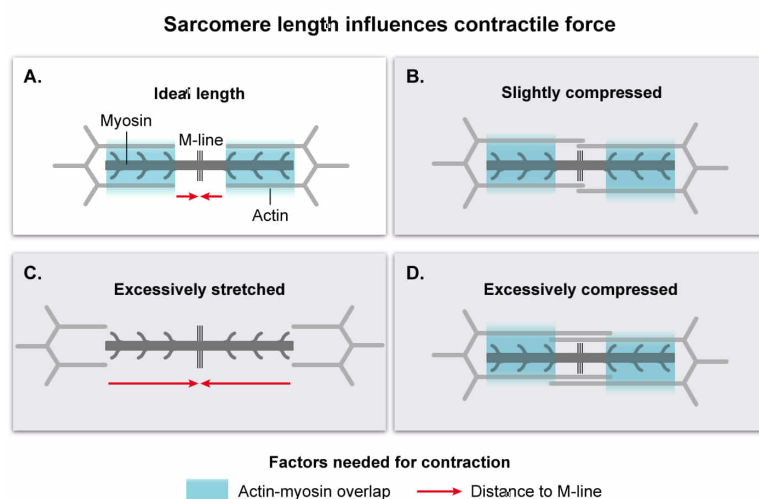


[14%]

Correct

50% answered correctly

Explanation:



Skeletal muscle, comprised of multinucleated muscle fiber cells, facilitates voluntary movement. Each muscle fiber contains thousands of individual functional units called **sarcomeres**. Sarcomeres consist of both **thin filaments (actin)** and **thick filaments (myosin)**, and **muscle contraction** depends on the interaction between these filaments. During skeletal muscle contraction, myosin heads bind the actin filaments and slide them toward the center of the sarcomere (ie, the M-line), shortening the sarcomere's overall length. When muscle relaxes, myosin heads release the actin filaments and the sarcomere returns to its original resting length.

The question states that the ability of a resting sarcomere to contract is dependent on the existence of overlap between actin

and myosin filaments and the presence of a gap between the ends of the actin filaments and the M-line. At the ideal sarcomere resting length, actin filaments must *overlap* with myosin filaments to allow **maximum binding of myosin heads to actin** on contraction. In addition, the distance between the ends of the actin filaments and the M-line should be *maximized* to provide the **greatest distance** for actin filaments to slide during contraction. Sarcomeres exhibiting these characteristics generate the **greatest** amount of **contractile force**.

(Choice C) The question states that contractile ability is dependent in part on an overlap between actin and myosin filaments. The sarcomere shown does not possess this overlap; in other words, it is excessively stretched. Therefore, it does *not* have the potential to generate the greatest amount of contractile force, and instead contractile force will be *reduced* in this sarcomere.

(Choices B and D) According to the question, contractile ability depends in part on the presence of a gap between the ends of the actin filaments and the M-line. The sarcomeres shown do not have this gap, which leads to a *decrease* in the amount of contractile force that can be generated by these sarcomeres.

Educational objective:

Skeletal muscle fibers consist of thousands of individual functional units called sarcomeres, and each sarcomere is comprised of both thin actin and thick myosin filaments. On muscle contraction, myosin heads bind the actin filaments and slide them toward the center of the sarcomere.

Time spent:0 sec

Subject:
Biology

QID:402683

System:
Musculoskeletal System